Software Defined Radio based Emulation of SAT-Terrestrial Network Coexistence

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***Abstract*—As cloud applications and 5G use cases increase the demand for additional spectrum, there has emerged interest in sharing the spectrum between terrestrial and satellite (SAT) networks. This paper assesses Dynamic Exclusion Zones (DEZ) and Power Control for spectrum coexistence between 5G Networks and SAT Digital Broadcasting Stations (DVB-S2). Relying on a baseband Software Defined Radio (SDR) based emulation on the COSMOS testbed, we propose a spectrum sharing control model that leverages a joint Dynamic Radio Resource Management (RRM) algorithm (Power Control with DEZ) for optimized resource management. Furthermore, the model combines these RRM strategies with artificially generated atmospheric attenuation and path losses for robust control in a practical communication environment. Evaluating our implementation with a case study of a Rural Macro region, we emphasize our system's effectiveness by accounting for power control strategies, preservation of DVB-S2 receiver's Packet Error Rate (PER), and the 5G network's PER minimization. Our results reveal that joint RRM strategies can restore link performance to within 73% to 99% of the ideal (no-interference) scenario under different weather conditions.**

***Index Terms*—Spectrum Sharing, Power Control, Dynamic Exclusion Zones, 5G**

I. Introduction

The development of commercial 5G New Radio (NR) applications has rampantly advanced and challenged the requirement for reliability, low latency, and spectrum utilization in wireless networks [1], [2]. As innovations in spectrum bands below 7.125 GHz for Frequency Range 1 (FR1) and in 24 to 52.6 GHz for Frequency Range 2 (FR2) face spectrum insufficiency and mmWave deployment challenges, the 7.125 to 24 GHz Frequency Range 3 (FR3) provides new opportunities for expansion with favorable radio propagation characteristics and enabling small cell Base Stations (BS) to expand 2.33 times past its FR2 coverage radius [3]. However, as active and passive satellite (SAT) systems occupy the FR3 band, the challenges of spectrum coexistence and mitigating interference from 5G highlight the necessity for spectrum sharing algorithms and Dynamic Radio Resource Management (RRM).

Therefore, it is crucial to evaluate a new dynamic, spectrum coexistence framework that accommodates multiple networks, prevents obstruction to SAT services' performance, and realizes the potential of FR3's environmental characteristics. In this paper, we propose a spectrum sharing control model that facilitates power adjustments via Software Defined Radio (SDR) based emulation, co-located sensor deployment, and intelligent feedback to handle coexistence of baseband SAT and 5G NR transmissions [4]. This model adjusts for the realistic atmospheric propagation effects in the FR3 band and 5G's Orthogonal Frequency Division Multiplexing (OFDM) waveforms, whose radio waves can be augmented by its mercurial environment.

Our Contributions: Based on these requirements, this paper makes the following contributions:

1) Emulating a 5G and SAT Digital Broadcasting Station (DVB-S2) interference scenario through SDR emulation
2) Creating a joint RRM (Dynamic Exclusion Zone and Power Control) technique to maximize SAT service quality and mitigate 5G interference
3) Assessment of system model performance through a case study and emulation on COSMOS [5] testbed

The paper is organized as follows: Section II provides Literature Review, Section III presents the System Model, Section IV explains the Emulation Framework, Section V describes the RRM techniques, Section VI details a Case Study regarding a Rural Macro interference scenario, Section VII presents Results, and Section VIII details Conclusions and Future Work.

II. Literature Review

Opportunities for modeling 5G's interference, SAT services, and efficient spectrum algorithms in the FR3 band have enabled efficient RRM research and 5G spectrum sharing surveys, as documented in the following studies [6] and [7]. However, several implementations encounter difficulties with environment adaptation, which is pivotal for FR3 modeling. Consequently, this review delves into various RRM simulations and implementations through the following subsections.

A. Dynamic Exclusion Zone Simulations

Offering a context-based spectrum sharing model, authors of [8] evaluate RRM policies such as Dynamic Exclusion Zones (DEZ), demonstrating a computer-simulated emulation of the 5G environment and accounting for network information. However, this implementation is limited by its immense computational infrastructure and virtualization, hindering scalability of different physical resource management strategies like transmission power. Similarly, DEZ research through Monte-Carlo simulations exemplifies formulations of DEZ policies

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in statewide SAT spectrum sharing [9], [10]; nevertheless, these emulations do not acknowledge hardware-based RRM strategies and generalize the parameters for nuanced FR3 communication environments. Our work explores these potential insights by employing a SDR based emulation, assessing the waveform and RRM techniques on adaptable hardware.

B. Weather Modeling

The construction of FR3 scenarios produces intriguing environmental frameworks, where the authors of [11] address FR3 atmospheric losses and path losses with prominent models. While their research proposes realistic multi-path and receiver gain losses, emulation constraints of 50% User Equipment (UE) loading and assumptions of the antenna do not address the DVB-S2 waveform interaction with the hardware. Additionally, they expand on their environmental framework with receiver antenna patterns and FR3 standardization consistent to 3GPP requirements [12]. Nevertheless, this approach only approximates environmental conditions through non-line-of-sight path losses, instead of factoring more diverse weather propagation effects or SAT downlink scenarios in FR3. Addressing FR3's environmental effects, we incorporate a diverse set of atmospheric and path losses in baseband processing, while reproducing the SAT downlink service on the SDR.

C. Spectrum Sharing, Spectrum Sensing, and Radio Resource Management

The field of dynamic spectrum access proposes advancements and frameworks that involve cognitive radio deployment, along with intelligent power control. Authors of [13] conceptualize a similar framework, leveraging significant computational resources and implementing a deep reinforcement learning approach. Comparatively, another research group involves sensor spectrum monitoring to assess leakage, regulating coexistence for sensitive multi cell TDD networks [14]. These approaches combine spectrum sensing and RRM through hardware, but the lack of emphasis on FR3's propagation characteristics, significant computational overhead, and limited software stack makes it inaccessible for models of SAT. Meanwhile, authors of [15] propose a channel allocation based power control policy and employing RRM to handle a BS's resources, but the implementation characterizes a minimalist indoor environment, which limits its utility for FR3 environments and other RRM solutions like DEZ. Physical setups found in a plethora of studies explore spectrum sharing through massive software infrastructures and complex sensor readings, which serve as real-life applications of spectrum sharing [16], [17]. Contrasting these systems, we leverage observations via SDR based Adaptive Energy Detection to reduce software complexity.

III. Spectrum Coexistence Model

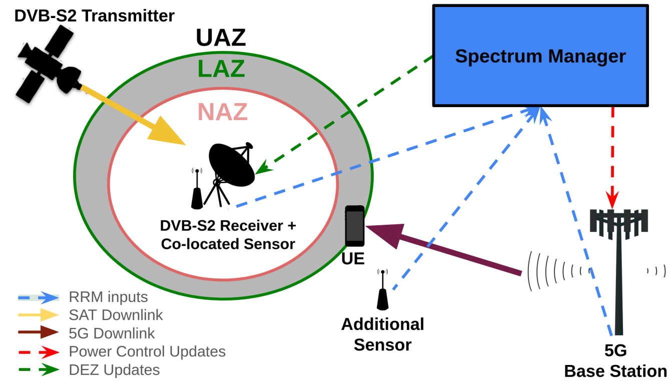


Fig. 1: Spectrum Management Model

As shown in Fig. 1, the spectrum coexistence model assesses co-located DVB-S2 and 5G links, where RRM control algorithms handle the transmission power and DEZ for the corresponding 5G BS. The model assumes two spectrum sensors, where one is co-located with the DVB-S2 receiver and the other is placed between the DVB-S2 receiver and 5G BS. These sensors provide capability for observations of Signal-to-Noise-Ratio (SNR) and energy in a specified channel, which can be inputs for RRM. Even though the emphasis is on a single BS, a single UE, and a single DVB-S2 link, this model can be scaled with an appropriate number of sensors for different interference mitigation and spectrum management scenarios of 5G and DVB-S2 networks.

As seen in Fig. 1, the 5G link is located in proximity to the DVB-S2 receiver's DEZ [18] of No Access Zone (NAZ), Limited Access Zone (LAZ), or Unlimited Access Zone (UAZ). These DEZ determine the 5G network's transmission power regulation in a 100% spectrum overlap scenario. Specifically, these DEZ expand in size, which affects the corresponding transmit power output from the 5G BS to the UE. The goal of the spectrum sharing model is to assess the RRM technique performance and spectrum coexistence enabled between the DVB-S2 and 5G links.

To enable coexistence, a spectrum manager shown in Fig. 1 takes inputs from the sensors, 5G BS, and measured PERs (Packet Error Rate) at the DVB-S2 receiver and UE. These inputs are utilized in designing robust RRM mechanisms that control both the DEZ and 5G transmitter power. It also takes into account the fluctuating environmental impacts associated with the FR3 band, due to different weather conditions.

IV. Emulation Framework

A. Interference Model

Establishing optimized RRM techniques for spectrum coexistence between 5G networks and SAT services in FR3 presents significant infrastructure and resource allocation challenges. As it is infeasible to test with active 5G infrastructure and to develop a fully-realized commercial satellite, we prototype the spectrum coexistence model with SDR based emulation of 5G OFDM and DVB-S2 protocols. While this emulation does not reflect real 5G BS or DVB-S2 transmissions, the SDRs' customization provides baseband processing and software accessibility to emulate these networks' interference characteristics. Furthermore, as deployment is confined to the size of the COSMOS testbed, the framework can only represent geographical alterations of the DEZ, through artificial variables and physical distances held in the COSMOS testbed. Thus, the effects of baseband processing and SDR attenuation will account for these deficiencies, replicating a proper geographic scenario [19], [20]. Furthermore, in the results presented later, we only consider the effect of the 5G BS downlink and its interference caused to the DVB-S2 downlink. Without loss of generality, the framework presented in the rest of the paper can be adapted to include the case of the uplinks as well.

B. Hardware and Software Setup

Experiments were performed in the COSMOS testbed, utilizing SDRs to incorporate RRM strategies. The B210 [21] USRPs were utilized for the 5G OFDM network and DVB-S2 reception. The satellite transmission required a X310 [22] SDR, along with B210s at the DVB-S2 co-located sensor and additional sensor between the 5G and DVB-S2 network for detection. Overall, the networks' software involved GNU Radio 3.10 [23] performing the baseband transmissions and operations, while on Linux Ubuntu 22.04 OSs.

C. 5G Waveform

Our model uses a GNU Radio based 64-bin OFDM, representing a stable and flexible waveform structure against frequency-selective fading. In addition, the OFDM reinforces 5G NR experimentation on narrowband interference in a dynamic spectrum context, permitting opportunity for scalable scenarios and waveform modifications. For specifications, the transmitter that emulated the 5G BS network is modulated by Quadrature Phase-Shift Keying for the header information, while a 16 Quadrature Amplitude Modulation is employed for the payload information. A sync word was utilized for the 5G network to eliminate synchronization mismatch.

D. Digital Video Broadcasting Waveform

We incorporate a DVB-S2 transmission through GNU Radio, which allowed variability in modulation and forward error correction. For handling potential interference, the employment of a physical layer frequency correction ensures the DVB-S2 protocol can handle time synchronization, forward error correction, and frame processing to ensure reliable SAT performance. Modulations for the DVB-S2 were conducted in 8 Phase-Shift Keying with $\frac{3}{5}$ low-density parity error correction on normal frames to match ETSI's standards. Emulating the video source, a MPEG Transport Stream will be transmitted, executing the transmission through a format utilized in current DVB-S2 protocols.

E. Weather Modeling

The FR3 weather modeling algorithm supplements baseband radio waves with artificial noise corresponding to path losses and FR3's atmospheric losses: cloud, snow, rain, oxygen, and water vapor. To approximate these losses, the usage of 3GPP [24] and ITU-R tables were implemented for FR3's atmospheric loss. Path loss (in dB) relies upon a myriad of factors that have to be accounted for in the the COSMOS testbed: no obstacles or buildings, extensive distance of d (meters) between transmitter and receiver, and a required emulation of an outdoors scenario. As path loss deals with d and frequency f_c of the given signal, the line-of-sight (LOS) path loss is derived from a modified equation [11]:

$$PL(d) = PL_{LOS}(f_c, d) + \mathcal{X}(0, \sigma_{LOS}). \tag{1}$$

Here, the path loss represents a LOS model derived from the 3GPP model [24], providing an equation with inputs of f_c and d for propagation loss. It also utilizes σ_{LOS}, which refers to the standard deviation of the environment's zero-mean shadow fading loss $\mathcal{X}$. As the COSMOS testbed emulation facilitates an open outdoors environment, the recommendations found in ([24], Table 7.4.1) provides sufficient parameters to calculate loss in both the 5G and SAT networks.

Accounting for the DVB-S2 downlink path loss to the receiver and 5G signal attenuation to UE, the following equation is utilized for attenuating transmission and adding relative atmospheric loss:

$$A_{Fall}(\theta, h, T, R, f_c) = (\mathbf{1}_a A_{env}(f_c, R))(A_{atm}(h, f_c))G(\theta),$$

where h, T, R, and f_c denote the attenuation's parameters of height (kilometers) in effective slant range, temperature (Kelvin), rain rate (mm/h), and baseband processing frequency. For $A_{env}(f_c, h)$, the attenuation value is produced by the associated frequency's ITU-R table, and $\alpha \in \{0, 1, 2, 3\}$ is a selection for either ideal, day, rain, or snow scenario, which multiplies with the general atmospheric attenuation $A_{atm}(h, f_c)$. For the ideal case, attenuation value excludes factors associated with clouds and humidity. $G(\theta)$ is provided to produce an imitated gain coinciding with elevation (θ) angle of a particular receiver and transmitter pair. To emulate the constraints of the DVB-S2 protocol, the model iterates through a designated downlink time, adjusting for weather conditions on a consistently updated slant range and $10° < \theta < 90°$ elevation angle. Thereby, the accuracy in modeling the DVB-S2 characteristics along with FR3 band can be achieved, and if there is additional environment scenarios, the calculations can be integrated into the iterative sequence of the DVB-S2 geostationary path model.

V. Radio Resource Management

A. Dynamic Exclusion Zones

The DEZ algorithm adjusts the 5G BS's transmission power in accordance to an artificial boundary. During the DVB-S2's downlink, this algorithm uses collected data at the DVB-S2 co-located sensor. As data acquisition enables the DEZ updates, the comprehension of the DEZ control requires data extrapolated by the DVB-S2 co-located sensor to make an assessment of the overall environment.

Upon registering an inadequate SNR value or error in the physical layer frequency correction, the receiver updates the DEZ zone in accordance to the LAZ and NAZ to deter the 5G BS. Through iterations, the process will maneuver through radius (Δr) kilometer updates, continuing the process until the 5G BS transmission is terminated or the DVB-S2 signal is fairly distinguishable. Based on improving SNR, the calculated shifts are the LAZ and NAZ radii updates, which determine 5G transmission power if the set radius threshold is surpassed.

For the radii assignments, it will be constructed in the following setup:

$$r_{BS} \geq \begin{cases} \Delta r_{NAZ} + r_{NAZ}(n) = r_{NAZ}(n+1) \\ \Delta r_{LAZ} + r_{LAZ}(n) = r_{LAZ}(n+1), \end{cases} \tag{2}$$

where Δr_{LAZ} and Δr_{NAZ} are the initial radii updates assigned upon initialization of the DEZ, r_{LAZ} and r_{NAZ} refer to the radii of the respective zone based on their proper n step increment, and the r_{BS} vector refers to the two element BS threshold that changes the 5G BS transmission output once eclipsed. As this implementation increases NAZ and LAZ, the following radii will either converge to the 5G BS location and prohibit transmission or guarantee a safe DVB-S2 receiver SNR, while the 5G transmission is limited.

B. Power Control

The co-located sensor utilizes Energy Detection across a bandwidth associated with the 5G BS communication, separating readings into 10 MHz channels to assess a 100% overlap scenario. As the sensor between the DVB-S2 receiver and 5G BS is severely affected by the 5G BS's overlap, calculated readings of Signal-to-Interference-Noise-Ratio (SINR) becomes inefficient, since the BS distorts the sensor's true signal power readings. Accounting for this, an implementation of Exponential Moving Average (EMA) is utilized. The formula accounts for E_{avg} or average energy in dB units by measuring overall energy of N channels in i iterations, multiplying it by weight μ to handle future updates, and averaging in context to recent value by accounting for time through t:

$$E_{avg} = \frac{1}{N}(\mu \sum_{i=1}^{N} |Y_t[n]|^2 + (1-\mu)\sum_{i=1}^{N} |Y_{t-1}[n]|^2). \tag{3}$$

Equation 3 describes the EMA observation input used in a Linear Quadratic Regulator (LQR) based power control algorithm to adapt to the interference. Specifically, the algorithm restricts E_{avg} from falling below a given threshold, which on the COSMOS testbed is set to a relative gain of -70 dB. While this implementation cannot distinguish between either network's signal, it provides a limit on the 5G BS transmission in order to reduce distortion on the DVB-S2 signal.

C. Joint Power Control with Dynamic Exclusion Zone

We present the joint RRM strategy that combines LQR based power control with DEZ, as shown in Algorithm 1. Specifically, Algorithm 1 contextualizes the spectrum sharing environment as a discrete-nonlinear system that imposes restrictions on 5G BS transmit power θ_p from [0,1.0]. The observations and the comprised state vector relies upon the SNR and energy values extrapolated from a sensor co-located with the DVB-S2 receiver (see Fig. 1), a locking mechanism to ensure SNR data are not erroneous, and measured PER associated with the 5G UE. Meanwhile, the control input $u[n]$ derives from modifying it by a calculation of K gain matrix. Effectively, this control input is then taken in context to a DEZ, where exceeding threshold of r_{BS} makes the control input a scalar for a limited LAZ transmission power of P_{LAZ} instead of full transmission power P_T. The LQR based power control incorporates weighted A (input) and B (input-to-state) matrices that highlight how power adjustment influences all elements of the state vector. To ensure preservation of the 5G BS link performance, the LQR places a higher penalty weight on PER for the UE to ensure the 5G link does not deteriorate.

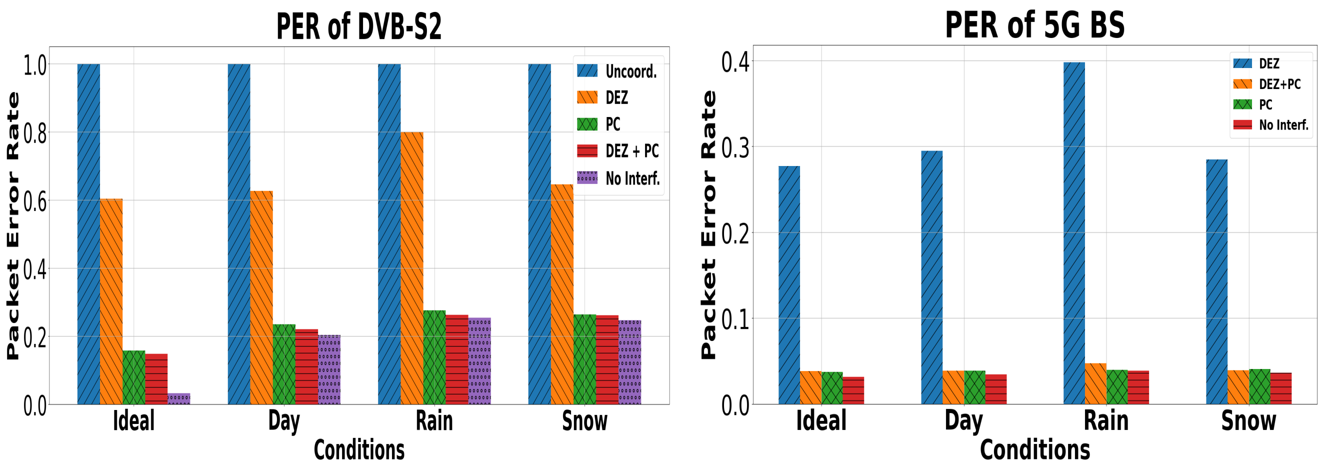
Nevertheless, if the observational sequence of the DVB-S2 receiver fails to report a finding, this indicates a lock on the mechanism, which is accounted for by the LQR's adjusted Q matrix whose entries are state cost penalty weights. If the DVB-S2 receiver is deemed locked in its given iteration, then the Q matrix enforces heavier costs on sensor energy readings and PER. Note that the LQR policy in Algorithm 1 employs a Discrete Algebraic Ricatti Equation (DARE).

Algorithm 1 LQR Control with Lock Condition, Power Constraints, and DEZ

$u[n] \leftarrow \max(0, \min(1.0, u[n]))$ # Initialize control vector
$\theta_p \leftarrow 1.0$; $u[0] \leftarrow \theta_p$ # Initial control input
$x[n] \leftarrow zeros()$ # Initial state vector
A, $B \leftarrow inputs()$ # Input and input-to-state matrices
Q_{normal}, $Q_{\text{lock}} \leftarrow states()$ # State cost penalty matrices
$R \leftarrow diag(1)$ # Set single input-cost matrix
while system is active **do**
  Measure state: $x[n] \leftarrow \{Energy, SNR, PER, lock\}$
  if $lock$ condition is true **then**
    $Q \leftarrow Q_{\text{lock}}$
  else
    $Q \leftarrow Q_{\text{normal}}$
  end if
  Solve P by solving $DARE(A, B, Q, R)$
  Compute LQR gain matrix: $K \leftarrow R^{-1}B^T P$
  Compute new control input: $u[n] \leftarrow -Kx[n]$
  Incorporate LAZ updates via Equation 2
  if $r_{LAZ}(n) \geq r_{BS}$ **then**
    Set $u[n] \leftarrow u[n] * P_{LAZ}$ # LAZ power threshold
  else
    Set $u[n] \leftarrow u[n] * P_T$ # UAZ power threshold
  end if
  Update system state on $u[n]$ and get feedback
end while

VI. Case Study: Rural Macro Region

We consider emulation of the DVB-S2 satellite's downlink baseband signal at the DVB-S2 receiver. We consider the effects of the FR3 environment by characterizing the downlink through atmospheric and path losses associated with 12.7 GHz. Even though these losses will be artificially generated, the transmission of the waveform will demonstrate a satellite pass



(a) Impact of RRM on DVB-S2

(b) Impact of RRM on 5G BS

Fig. 2: Spectrum Sharing between DVB-S2 and 5G BS under different weather conditions

in a 100% overlapping interference scenario. For the satellite, a SDR [22] will emulate a 10 minute downlink pass, relaying a MPEG transport stream that can be viewed by the receiver, and another SDR [21] will incorporate a baseband interpolation of the satellite's effective path as the elevation angle θ transitions from $0°$ to $90°$. The baseband processing will account for the distance between the SAT transmitter and receiver, by incorporating additional attenuation on the downlink signal.

The 5G terrestrial network includes a 5G BS communicating with 1 UE, where both BS and UE are emulated using SDRs on the COSMOS testbed. Using the radio mapping feature of the COSMOS Grid [19], [20], the link between the BS and DVB-S2 receiver is emulated in accordance with the Rural Macro model proposed by the authors of [8] and shown in Fig. 3, where the link distance is 10 km and includes additional parameters such as humidity, temperature, water vapor density, and rain rate. The rain rate of 25 mm/h parameter is modeled with parameters associated to an average day at sea level. The BS is located 1 km from the UE to be consistent with the Rural Macro setting. The only interference encountered by the DVB-S2 link is from the 5G BS transmission. The Rural Macro model has LOS path losses and does not include any evaluation of multi-path losses. However, it does include losses due to Rician fading to characterize the open rural area.

For implementing RRM strategies, the emulation utilizes another B210 SDR sensor located between the 5G BS and DVB-S2 receiver, as well as a centralized computer (spectrum manager) that aggregates data corresponding to each of the links' received signals. For the LAZ and NAZ, we initialize the radii to be 1 km and 0.5 km respectively. The sequential updates to these distances are based on the SNR values of the DVB-S2 co-located sensor and follow iterative increments of 1 km and 0.5 km respectively. If the LAZ is approached, the 5G BS transmission gain will be limited to ensure the SNR of the DVB-S2 receiver is an adequate signal of around 9.0 dB. Through aggregation of the data and utilization of post-processing of the waveforms, the RRM strategies will iterate in parallel to the SDRs' operations, performing real-time analysis and transmitter power control. To assess the performance, PER for the DVB-S2 receiver and the UE are measured.

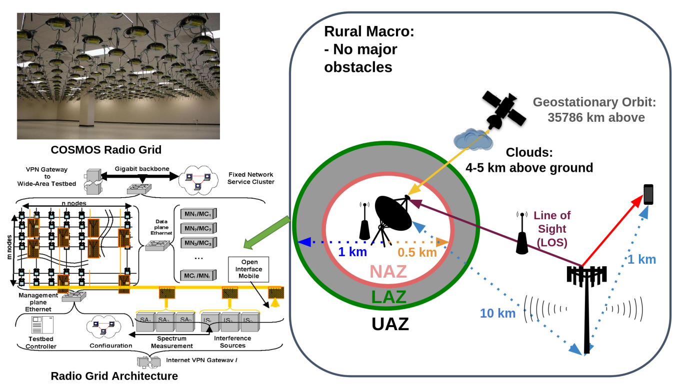


Fig. 3: Rural Macro Setup on the COSMOS Grid

VII. Results

Fig. 2(a) and 2(b) show the PER for the DVB-S2 and 5G BS links respectively for different weather conditions (Ideal, Day, Rain, Snow). Fig. 2(a) also shows the relative performance of each of the RRM techniques (DEZ, power control [PC], joint power control [DEZ + PC]), as well as uncoordinated spectrum sharing (Uncoord.) and no interference (No Interf.) scenarios for the DVB-S2 and 5G BS link. As an observation, the attenuation by any environment other than ideal results in a significant degradation of DVB-S2 performance. Even the no interference scenario results in a PER of 20%, due to the attenuation of day, snow, or rain preventing the SAT downlink from being accessible below an elevation of $20°$. The RRM technique of DEZ marginally improves performance, while the power control to the 5G BS transmission significantly improves DVB-S2 performance. Specifically, the implementation of power control for DVB-S2 provides a nearly 40% PER improvement to that of DEZ regardless of weather conditions. Moreover, joint DEZ and power control almost restores PER performance close to that of the no interference scenario. This highlights that the general FR3 spectrum sharing environment encounters significant depreciation of the DVB-S2 signal, emphasizing the need for RRM techniques in 5G terrestrial networks to preserve the requisite fidelity. Figure 2(b) shows the impact of such RRM for spectrum coexistence on the 5G BS link. It is observed that once again power control greatly improves the PER performance of the 5G BS link for all weather conditions. In summary, it is observed that joint RRM restores the PER performance for both 5G BS and DVB-S2 links to that of the no interference scenario.

VIII. Conclusion and Future Work

This paper demonstrated the capability of joint RRM techniques including LQR based power control and DEZ for DVB-S2 and 5G networks through the use of the SDR based COSMOS testbed. Showcasing reliable results in the attenuated environments, joint RRM was fundamental in preserving the DVB-S2's PER, while ensuring 5G BS PER minimization. Relying on the co-located and additional sensor, the observational data obtained from the SDRs could provide more insight into complex FR3 environments, while scaling for the inclusion of further interference, additional 5G BS, and other optimized approaches in reinforcement learning.

As the COSMOS testbed relies upon the usage of SDRs, FR3 experimentation is only tangible through the modality of the SDRs used in emulation. To improve upon the given SDR based emulation, future experiments will utilize a Transceiver Design that upconverts the SDRs' transmissions to 12 GHz. Through this design, the SDR has flexibility for higher frequencies and more distinct path loss modeling [25]. Therefore, the use of the SDR and this transceiver design can provide a versatile set of applications and reliable modeling, which can be incorporated into the future considerations of the SDR based Emulation between SAT and 5G BS.

IX. Acknowledgements

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